

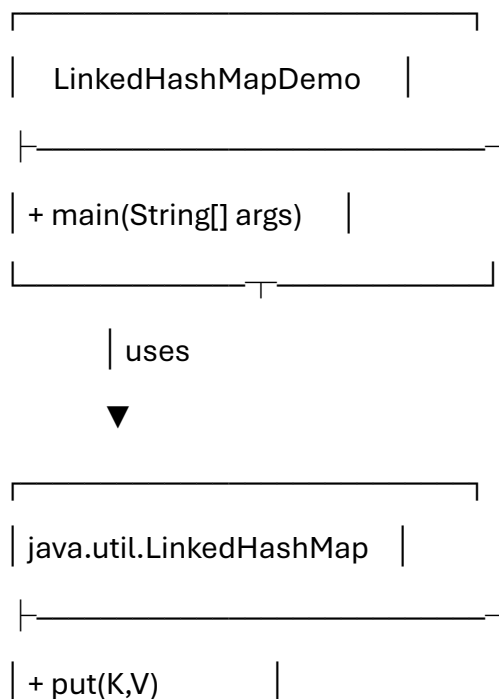
Aim

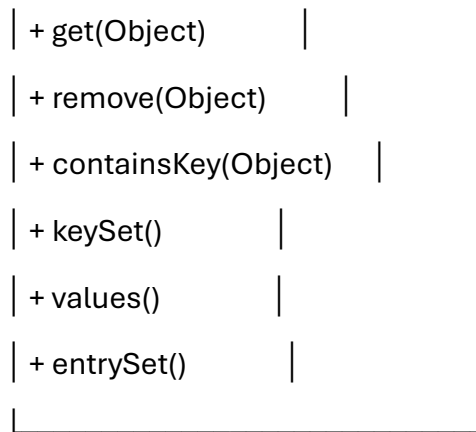
To write a Java program to demonstrate the important methods of the **LinkedHashMap** class, such as put(), get(), remove(), containsKey(), keySet(), values(), and entrySet().

Algorithm

1. Start the program.
2. Import the java.util.LinkedHashMap class.
3. Create a LinkedHashMap object.
4. Add key-value pairs using put().
5. Retrieve a value using get().
6. Check whether a key exists using containsKey().
7. Remove a key-value pair using remove().
8. Display all keys using keySet().
9. Display all values using values().
10. Display all key-value pairs using entrySet().
11. Display the results.
12. Stop the program.

UML Diagram





Code:

```
package Observations;

import java.util.LinkedHashMap;

public class LinkedHashMapClass {

    public static void main(String[] args) {

        // Creating LinkedHashMap
        LinkedHashMap<Integer, String> map = new LinkedHashMap<>();

        // put()
        map.put(101, "Apple");
        map.put(102, "Banana");
        map.put(103, "Mango");
        map.put(104, "Orange");

        System.out.println("LinkedHashMap: " + map);

        // get()
        System.out.println("Value for key 102: " + map.get(102));
```

```

// containsKey()

System.out.println("Contains key 103: "
    + map.containsKey(103));

// remove()

map.remove(104);

System.out.println("After removing key 104: " + map);

// keySet()

System.out.println("Keys: " + map.keySet());

// values()

System.out.println("Values: " + map.values());

// entrySet()

System.out.println("Key-Value pairs: " + map.entrySet());
}
}

```

Output:

```

es -cp C:\Users\Sahithi\pyla\AppData\Roaming\Code\User\Workspaces
bservations.LinkedHashMapClass'
LinkedHashMap: {101=Apple, 102=Banana, 103=Mango, 104=Orange}
Value for key 102: Banana
Contains key 103: true
After removing key 104: {101=Apple, 102=Banana, 103=Mango}
Keys: [101, 102, 103]
Values: [Apple, Banana, Mango]
Key-Value pairs: [101=Apple, 102=Banana, 103=Mango]
PS C:\Users\Sahithi\pyla\Desktop\Sahithi java>

```

Result:

Thus, the Java program successfully demonstrates the LinkedHashMap methods put(), get(), remove(), containsKey(), keySet(), values(), and entrySet(), while maintaining the insertion order of key-value pairs.